Session 6: Make the numbers talk

Question: How do we show the finding so a manager understands it in five seconds?

Concepts

Concept	In daily life	In Excel	The limit
merge	Matching a packing list to the catalogue	VLOOKUP, XLOOKUP	Count rows before and after. A repeated key makes the table grow
matplotlib	Drawing a chart by hand, step by step	Insert Chart, then edit each part	Every chart needs a title and axis labels
seaborn	A chart template on top of matplotlib	Chart styles	Titles and labels work the same way

Code patterns

```
print(len(product_value))
product_value = product_value.merge(names, on="stock_code", how="left")
print(len(product_value))   # same number?


plt.figure(figsize=(10, 5))
plt.barh(top10["description"], top10["revenue"])
plt.title("The finding, in words")
plt.xlabel("Revenue (pounds)")
plt.ylabel("Product")
plt.tight_layout()
plt.show()

plt.xticks(rotation=45)   # when labels are long
```

Three common errors

Error	What to do
Table has more rows after a merge	A key repeats on the right. Use drop_duplicates(subset="stock_code") on the names table.
Chart shows twice	You called plt.show() two times. Call it once at the end.
Labels are cut off	Add plt.tight_layout() before plt.show().

Exercises and expected answers

Exercise	Expected
Predict: After this merge, will the table have more, fewer or the same 3,066 rows? Guess first.	Guess first, then run it to check
Fill in: Show the top 3 products with their names.	REGENCY CAKESTAND 3 TIER on top
Fix it: This chart has no title and no labels. Add a title that says the finding, and both axis labels.	The cell runs without an error
Fix it: One column name is wrong. Fix it so the chart draws.	The cell runs without an error
Fill in: Draw the daily line chart with a finding title and two axis labels.	Your own
Predict: Which day had the most revenue? Guess from the chart first.	Guess first, then run it to check
Write it: The revenue on the best day, rounded to 2 decimals.	81454.99
Write it: A bar chart of revenue per month, with a finding title and axis labels.	three bars, December the tallest
Case: A sales manager wants to visit one country outside the UK. Show the top 5 countries outside the UK in a bar chart with a finding title, and print the top one.	Netherlands 58182.55
Extra: The same top 5 countries with seaborn. Use sns.barplot with x and y.	Your own
Extra: The 80% curve: plot cumulative_share against the product rank, with a dashed line at 0.80 (plt.axhline(0.80, linestyle="--")).	Your own

Case hunt: one page, five points

Point	Your answer
1. The problem	
2. Who is affected	
3. What data would show it	

4. One question you could answer with data	
5. Which tool fits: Python, SQL or a dashboard	